

$$\begin{aligned}
& \frac{1}{16\pi^2} \left( \frac{4}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_d^p] \delta_{i1i2} - \frac{5}{54} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_d^p] \delta_{i1i2} + \frac{16}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_d^p] \delta_{i1i2} + \right. \\
& \frac{27}{4} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_e^p] \delta_{i1i2} - \frac{18}{5} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_e^p] \delta_{i1i2} + \frac{16}{135} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_e^p] \delta_{i1i2} + \\
& \frac{2}{27} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_l^p] \delta_{i1i2} - \frac{5}{36} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_l^p] \delta_{i1i2} + \frac{8}{135} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_l^p] \delta_{i1i2} + \\
& \frac{2}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_q^p] \delta_{i1i2} - \frac{5}{108} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_q^p] \delta_{i1i2} + \frac{8}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_q^p] \delta_{i1i2} + \\
& \frac{16}{81} \sum_p \mathbf{g}_1^4 \text{LF}_{3,0}[\mathbf{m}_u^p] \delta_{i1i2} - \frac{10}{27} \sum_p \mathbf{g}_1^4 \text{LF}_{4,-1}[\mathbf{m}_u^p] \delta_{i1i2} + \frac{64}{405} \sum_p \mathbf{g}_1^4 \text{LF}_{5,-2}[\mathbf{m}_u^p] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} + \frac{1}{9} \mathbf{g}_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} - \frac{16}{135} \mathbf{g}_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} + \\
& \frac{1}{36} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_q^p] - \frac{1}{18} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_q^p] + \\
& \frac{1}{36} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_q^p] + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} - \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \\
& \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,1,0}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{2,2,-1}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} - \\
& \frac{4}{81} \mathbf{g}_1^4 \text{LF}_{3,1,-1}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \frac{2}{81} \mathbf{g}_1^4 \text{LF}_{4,1,-2}[\mathbf{m}_1, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \\
& \frac{3}{4} \mathbf{g}_2^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{2,1,0}[\mathbf{m}_2, \mathbf{m}_q^p] - \frac{3}{2} \mathbf{g}_2^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{3,1,-1}[\mathbf{m}_2, \mathbf{m}_q^p] + \\
& \frac{3}{4} \mathbf{g}_2^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{4,1,-2}[\mathbf{m}_2, \mathbf{m}_q^p] + \frac{4}{3} \mathbf{g}_3^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{2,1,0}[\mathbf{m}_3, \mathbf{m}_q^p] - \\
& \frac{8}{3} \mathbf{g}_3^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{3,1,-1}[\mathbf{m}_3, \mathbf{m}_q^p] + \frac{4}{3} \mathbf{g}_3^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{4,1,-2}[\mathbf{m}_3, \mathbf{m}_q^p] + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,2,-1}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} - \\
& \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{4,1,-2}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}1}] \delta_{i1i2} + \\
& \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,1,0}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{2,2,-1}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} - \\
& \frac{4}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{3,1,-1}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \frac{2}{27} \mathbf{g}_1^2 \mathbf{g}_3^2 \text{LF}_{4,1,-2}[\mathbf{m}_3, \mathbf{m}_u^{\text{i}2}] \delta_{i1i2} + \\
& \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_d^r, \mathbf{m}_q^p] \delta_{i1i2} - \frac{1}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] \delta_{i1i2} + \\
& \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_e^r, \mathbf{m}_l^p] \delta_{i1i2} - \frac{1}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_l^p] \delta_{i1i2} - \\
& \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_e^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] \delta_{i1i2} + \\
& \frac{1}{2} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_e^{\text{pr}} y_e^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_l^p, \mathbf{m}_e^r] \delta_{i1i2} - \\
& \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] \delta_{i1i2} - \frac{2}{9} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] \delta_{i1i2} + \\
& \frac{5}{6} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] \delta_{i1i2} - \frac{2}{3} \mathbf{g}_1^2 \tilde{\mu}^2 \overline{y}_d^{\text{pr}} y_d^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] \delta_{i1i2} - \\
& \frac{1}{9} \mathbf{g}_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] \delta_{i1i2} + \frac{1}{18} \mathbf{g}_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] \delta_{i1i2} - \\
& \frac{1}{18} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{2,1,0}[\mathbf{m}_q^p, \tilde{\mu}] + \frac{1}{36} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{2,2,-1}[\mathbf{m}_q^p, \tilde{\mu}] + \\
& \frac{1}{36} \mathbf{g}_1^2 \overline{y}_u^{\text{pi}1} y_u^{\text{pi}2} \text{LF}_{3,1,-1}[\mathbf{m}_q^p, \tilde{\mu}] + \frac{1}{9} \mathbf{g}_1^2 \overline{a}_u^{\text{pr}} a_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_u^r, \mathbf$$